Property valuation agent — Evaluation Metrics

1. Overview

An Automated Valuation Model (PVM property valuation agent) estimates the value of a property using available property information, historical transactions, comparable properties, location data, and other relevant features.

To evaluate the performance of a PVM property valuation agent, it is important to measure not only how accurate the valuation is, but also how frequently the system is able to produce a valuation and how much uncertainty exists around its estimates.

The following metrics can be used to evaluate the PVM property valuation agent:

- PPE10
- PPE20
- MdAPE (Median Absolute Percentage Error)
- FSD (Forecast Standard Deviation)
- Coverage Rate

2. PPE10

Definition

PPE10 measures the percentage of property valuations that are within ±10% of the actual transaction price. It answers the question: "How many of the model's valuations are within 10% of the actual selling price?"

Calculation

For each property:
Absolute Percentagge Error = |Predicted Value – Actual Value| / Actual Value × 10

0

A prediction is considered a PPE10 success if the absolute percentage error is 10% or less.

Example

Suppose:

- Actual transaction price = ₹50 lakh
- PVM property valuation agent predicted value = ₹46 lakh

The error is: |46 – 50| / 50 × 100 = 8%
Since 8% is less than 10%, this property is counted as a PPE10 success.

If 70 out of 100 properties have errors of 10% or less:
PPE10 = 70%

Interpretation

A higher PPE10 indicates that a larger proportion of valuations are very close to the actual transaction price.

3. PPE20

Definition

PPE20 measures the percentage of property valuations that are within ±20% of the actual transaction price. It answers: "How many of the model's valuations are within 20% of the actual selling price?"

Calculation

The absolute percentage error is calculated for each property. A prediction is counted as a PPE20 success if:
Absolute Percentage Error ≤ 20%

Example

Suppose 85 out of 100 property valuations have an absolute percentage error of 20% or less.
PPE20 = 85%

Interpretation

PPE20 provides a broader measure of valuation accuracy than PPE10.
For example:

- PPE10 = 60%
- PPE20 = 85%

This means 60% of valuations were within 10% of the actual price, while 85% were within 20%.

4. MdAPE — Median Absolute Percentage Error

Definition

MdAPE stands for Median Absolute Percentage Error. It represents the median of the absolute percentage errors across all evaluated properties. It answers: "What is the typical percentage error of the AVM?"

Calculation

First, calculate the absolute percentage error for every property:
Absolute Percentage Error = |Predicted Value – Actual Value| / Actual Value × 100

Then sort all errors from smallest to largest and take the median.

Example

Suppose the errors for five properties are: 4%, 10%, 25%, 8%, 5%
After sorting: 4%, 5%, 8%, 10%, 25%
The median is: MdAPE = 8%
This means that the median valuation error is 8%.

Interpretation

A lower MdAPE indicates lower typical valuation error.

5. FSD — Forecast Standard Deviation

Definition

FSD (Forecast Standard Deviation) represents the level of uncertainty associated with the PVM property valuation agent's valuation estimates. Unlike PPE10, PPE20, and MdAPE, which primarily measure the accuracy of predictions against actual transaction prices, FSD is intended to describe the variability or uncertainty associated with the forecasts.

Example

A PVM property valuation agent may produce:

- Estimated Property Value: ₹50 lakh
- Estimated uncertainty: ±₹5 lakh

The valuation can therefore be presented as an estimate around ₹50 lakh rather than as an exact value.

Importance

FSD can be useful when communicating valuation uncertainty to users. Two properties might both receive an estimated value of ₹50 lakh, but the model may have:

- Property A: low uncertainty
- Property B: high uncertainty

This indicates that the model has greater confidence in one valuation than the other.

Important Note

The exact calculation and interpretation of FSD should be defined according to the PVM property valuation agent's underlying statistical methodology. It should not automatically be treated as the standard deviation of raw prediction errors unless that is how the model has been designed.

6. Coverage Rate

Definition

Coverage Rate measures the percentage of property applications for which the PVM property valuation agent is able to produce a valuation. It answers: "For how many properties can the system actually generate a valuation?"

Calculation

Coverage Rate = (Number of properties with a successful valuation / Total number of property applications) × 100

Example

Suppose the system receives 1,000 property valuation requests. The PVM property valuation agent successfully generates valuations for 700 properties.
Coverage Rate: (700 / 1,000) × 100 = 70%
Therefore: Coverage Rate = 70%

Why Coverage Rate is Important

A model can have good accuracy on the properties it successfully values while still having limited overall coverage.
For example:

- 90% PPE20
- 10% MdAPE
- 45% Coverage

This means the model may be reasonably accurate when it produces a valuation, but it is currently able to produce valuations for only 45% of applications.
Therefore, coverage should be reported alongside accuracy metrics.

7. Recommended PVM property valuation agent Evaluation Report

The metrics can be presented together in a single evaluation table.

Metric	What it measures	Example
PPE10	% of valuations within ±10% of actual price	65%
PPE20	% of valuations within ±20% of actual price	87%
MdAPE	Median absolute percentage error	9.5%
FSD	Forecast uncertainty / variability	Model-dependent

Metric	What it measures	Example
Coverage Rate	% of applications for which a valuation was produced	72%

These metrics provide different perspectives on AVM performance:

- **PPE10 and PPE20** measure how frequently predictions fall within defined accuracy ranges.
- **MdAPE** describes the typical magnitude of valuation error.
- **FSD** describes forecast uncertainty according to the model's uncertainty methodology.
- **Coverage Rate** measures the proportion of applications for which the PVM property valuation agent can actually produce a valuation.

Together, these metrics provide a more complete evaluation of an Automated Valuation Model than using a single accuracy metric.